

Proof. We will use the feedforward block to calculate $k^2, X_{v,d}^2, v \in \cup I_1 \cup I_2$ as in the proof of Lemmas C.13 and C.14.

We then choose $W^{(K)}, W^{(Q)}$ such that

$$\begin{aligned} & \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} \\ &= -n^{4c+1} \sum_{i=1}^{|I|} (X_{I_1[i],k'} - X_{I_2[i],k})^2 - nk'^2 \\ & \quad + \mathbf{1}[k' = 1] \left(n^{4c+1} \sum_{i=1}^{|I|} X_{I_2[i],k}^2 + n - n^{2c+2} \right). \end{aligned}$$

The detailed construction of $W^{(K)}, W^{(Q)}$ is omitted here since it is similar to the proof of Lemmas C.13 and C.14.

We will discuss several cases for the distribution of $\left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k}$. It always holds that $\left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{1,k} = -n^{2c+2}$.

1. If there doesn't exist k' , such that $X_{k',I_1} = X_{k,I_2}$, then for any $i > 1$, we have $\left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{i,k} < -n^{4c+1}$.
2. If there exists k' , such that $X_{k',I_1} = X_{k,I_2}$, then for such k' , we have $\left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} = -nk'^2 > -n^{2c+1}$. The rest of the entries are all smaller than $-n^{4c+1}$. Finally, the largest k' satisfying that $X_{k',I_1} = X_{k,I_2}$ will correspond to a $\left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k}$ that is at least n larger than the second largest $\left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k}$, as in the proof of Lemma C.13.

Concluding the above discussion, we have after the column-wise softmax and rounding to $p = O(\log n)$ bit,

$$\left(\text{softmax} \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right) \right)_{k',k} = \begin{cases} \mathbf{1}[k' = \min\{h \mid X_{I_1,h} = X_{I_2,k}\}], & \{h \mid X_{I_1,h} = X_{I_2,k}\} \neq \emptyset \\ \mathbf{1}[k' = 1], & \text{otherwise} \end{cases}$$

Further, we will choose $W^{(V)}$ such that

$$\begin{aligned} W^{(V)} X'_{I_3,k'} &= X'_{I_4,k'} = X_{I_4,k'} \quad \forall k' \in [l]. \\ W^{(V)} X'_{I_3^c,k'} &= 0 \quad \forall k' \in [l]. \end{aligned}$$

This then concludes that

$$\mathcal{A}(g(X) + X) = \text{Match}[I_1, I_2, I_3, I_4](X)$$

This concludes the proof. $\square$

Lemma C.16 (Matching Closest Blocks). *Given any constant c , for index set $I_1, I_2, I_3, I_4 \subset [d-20], |I_1| = |I_2| \leq 10, |I_3| = |I_4|$, a matching closest function $\text{MatchClose}[I_1, I_2, I_3, I_4]$ can be implemented with 1 feedforward block g and 1 attention block $\mathcal{A}$ with 1 attention head. Formally, when $X_{d,k} = k, X_{3,k} = \mathbf{1}[k = 1], X_{I_1,1} = 0$ and $k \leq n^c$, it holds that*

$$\mathcal{A}(g(X) + X) = \text{MatchClose}[I_1, I_2, I_3, I_4](X)$$

Proof. The proof is similar to the proof of Lemma C.15, and one can design the attention pattern as

$$\begin{aligned} & \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} \\ &= -n^{4c+1} \sum_{i=1}^{|I|} (X_{I_1[i],k'} - X_{I_2[i],k})^2 - n(k - k')^2 \\ &+ \mathbf{1}[k' = 1] \left(n^{4c+1} \sum_{i=1}^{|I|} X_{I_2[i],k}^2 + n(k - 1)^2 - n^{2c+2} \right). \end{aligned}$$

The rest of the proof is omitted here. $\square$

Lemma C.17 (Matching Nearest Blocks). *Given any constant c , for index set $I_1, I_2, I_3, I_4 \subset [d - 20]$, $|I_1| = |I_2| \leq 10$, $|I_3| = |I_4|$ and index i , a matching nearest function $\text{MatchNearest}[I_1, I_2, I_3, I_4, i]$ can be implemented with 1 feedforward block g and 1 attention block $\mathcal{A}$ with 1 attention head. Formally, when $X_{d,k} = k$, $X_{3,k} = \mathbf{1}[k = 1]$, $X_{I_1,1} = 0$ and $k \leq n^c$, it holds that*

$$\mathcal{A}(g(X) + X) = \text{MatchNearest}[I_1, I_2, I_3, I_4, i](X)$$

Proof. The proof is similar to the proof of Lemma C.15, and one can design the attention pattern as

$$\begin{aligned} & \left(\left(W^{(K)} X \right)^\top \left(W^{(Q)} X \right) \right)_{k',k} \\ &= -n^{4c+1} \sum_{u=1}^{|I|} (X_{I_1[u],k'} - X_{I_2[u],k})^2 - n(X_{i,k} - k')^2 \\ &+ \mathbf{1}[k' = 1] \left(n^{4c+1} \sum_{u=1}^{|I|} X_{I_2[u],k}^2 + n(1 - X_{i,k})^2 - n^{2c+2} \right). \end{aligned}$$

The rest of the proof is omitted here. $\square$

Lemma C.18 (Matching Next Blocks). *Given any constant A, c , for index set $I_1, I_2, I_3, I_4 \subset [d - 20]$, $|I_1| = |I_2| \leq 10$, $|I_3| = |I_4|$ and a special counting index a , a matching next function $\text{MatchNext}[I_1, I_2, I_3, I_4, a]$ can be implemented with 1 feedforward block g and 1 attention block $\mathcal{A}$ with 1 attention head. Formally, when $X_{d,k} = k$, $X_{3,k} = \mathbf{1}[k = 1]$, $X_{I_1,1} = 0$ and $k \leq n^c$, it holds that*

$$\mathcal{A}(g(X) + X) = \text{MatchNext}[I_1, I_2, I_3, I_4, a](X)$$

Proof. We will use the feedforward block to calculate the following $\overline{\text{next}}$ function, where

$$\overline{\text{next}}(x) = \begin{cases} \frac{1}{2}, & x \geq \frac{2}{3}. \\ \frac{1}{3}, & \frac{3}{5} > x > \frac{2}{5}. \\ \frac{1}{4}, & \frac{7}{20} > x > \frac{3}{10} \\ x - x^2 + x^3, & x \leq \frac{11}{40}. \end{cases}$$

The value can be arbitrary for $x \in [\frac{11}{40}, \frac{3}{10}] \cup [\frac{2}{5}, \frac{7}{20}] \cup [\frac{3}{5}, \frac{2}{3}]$. This function is achievable by a feedforward block through combination of Lemmas C.1 and C.6.

The purpose of this is to approximate the next function for $x \in \text{ROUND}(1/[n^A], p)$, and we have the following lemma.

Lemma C.19. *For large enough n and any $x \in \text{ROUND}(1/[n^A], p)$, we have*

$$|\overline{\text{next}}(x) - \text{next}(x)| \leq \text{next}(x)^3 + O\left(\frac{1}{n^{10A}}\right).$$

Proof. We always have $\text{ROUND}(1/[n^A], p) \cap ([\frac{11}{40}, \frac{3}{10}] \cup [\frac{2}{5}, \frac{7}{20}] \cup [\frac{3}{5}, \frac{2}{3}]) = \emptyset$. We will discuss several cases for $x \in \text{ROUND}(1/[n^A], p)$.

1. If $x \geq \frac{3}{10}$, then $\overline{\text{next}}(x) = \text{next}(x)$.
2. If $x \leq \frac{7}{20}$, it holds that $|x - 1/m| \leq 1/n^{10A}$, $m \geq 3$, then

$$\begin{aligned}\overline{\text{next}}(x) &= x - x^2 = \frac{1}{m} - \frac{1}{m^2} + \frac{1}{m^3} + O\left(\frac{1}{n^{10A}}\right) \\ &= \frac{1}{m+1} - \frac{1}{m^3(m+1)} + O\left(\frac{1}{n^{10A}}\right)\end{aligned}$$

This then concludes the proof. $\square$

We then choose $W^{(K)}, W^{(Q)}$ such that

$$\begin{aligned}& \left((W^{(K)}X)^\top (W^{(Q)}X) \right)_{k',k} \\ &= -n^{4A+3} \sum_{i=1}^{|I|} (X_{I_1[i],k'} - X_{I_2[i],k})^2 - n^{4A+1} (\overline{\text{next}}(X_{a,k}) - X_{a,k'})^2 \\ & \quad + \mathbf{1}[k' = 1] \left(n^{4A+3} \sum_{i=1}^{|I|} X_{I_2[i],k}^2 + n^{4A+1} X_{a,k}^2 - n^{4A+2} \right).\end{aligned}$$

Again, the detailed construction of $W^{(K)}, W^{(Q)}$ is omitted here since it is similar to the proof of Lemmas C.13 and C.14.

We will discuss several cases for the distribution of $\left((W^{(K)}X)^\top (W^{(Q)}X) \right)_{k',k}$. It always holds that $\left((W^{(K)}X)^\top (W^{(Q)}X) \right)_{1,k} = -n^{4A+2}$.

1. If there doesn't exist k' , such that $X_{k',I_1} = X_{k,I_2}$, then for any $i > 1$, we have $\left((W^{(K)}X)^\top (W^{(Q)}X) \right)_{i,k} < -n^{4A+3}$.
2. If there exists k' , such that $X_{k',I_1} = X_{k,I_2}$, then for such k' , we have $\left((W^{(K)}X)^\top (W^{(Q)}X) \right)_{k',k} = -n^{3A}(\text{next}(X_{a,k}) - X_{a,k'})^2 > -n^{4A+1}$. The rest of the entries are all smaller than $-n^{4A+2}$.

It remains to discuss the distribution of $\left((W^{(K)}X)^\top (W^{(Q)}X) \right)_{k',k}$ for k' satisfying $X_{k',I_1} = X_{k,I_2}$. When X satisfies the condition in Definition C.12, we have that $S_k = \{X_{a,k'} \mid X_{k',I_1} = X_{k,I_2}\}$ takes consecutive and disjoint values in $\text{ROUND}(1/[n^A], p)$. Hence, if $|S_k| > 2$, suppose $y, z \in S_k$ satisfies that

$$\begin{aligned}|y - \overline{\text{next}}(X_{a,k})| &= \min_{x \in S_k} |x - \overline{\text{next}}(X_{a,k})| \\ |z - \overline{\text{next}}(X_{a,k})| &= \min_{x \in S_k, x \neq y} |x - \overline{\text{next}}(X_{a,k})|.\end{aligned}$$

We will discuss several cases for y, z .

- If $y - \overline{\text{next}}(X_{a,k})$ and $z - \overline{\text{next}}(X_{a,k})$ are both negative, then $y > z$, we have,

$$\begin{aligned}(y - \overline{\text{next}}(X_{a,k}))^2 - (z - \overline{\text{next}}(X_{a,k}))^2 &= (y - z)(y + z - 2\overline{\text{next}}(X_{a,k})) \\ &\leq -(y - z)^2 \leq -\frac{1}{4n^{4A}}.\end{aligned}$$

- If $y - \overline{\text{next}}(X_{a,k})$ and $z - \overline{\text{next}}(X_{a,k})$ are both positive, then $y < z$, and same as above we have

$$\begin{aligned}(y - \overline{\text{next}}(X_{a,k}))^2 - (z - \overline{\text{next}}(X_{a,k}))^2 &= (y - z)(y + z - 2\overline{\text{next}}(X_{a,k})) \\ &\leq -(y - z)^2 \leq -\frac{1}{4n^{4A}}.\end{aligned}$$